



Rishab Singh

+91 7668683441 — rishabsingh9897@gmail.com — <https://github.com/RishabSingh9897>

Summary — Electronics Engineering student passionate about **VLSI, semiconductor technology, and circuit design**. Interested in **space technology, astrophysics, and the physics behind electronic systems**, with a focus on applying electronics in space and communication technologies.

Education

Class X Passed with 93%

Class XII Passed with 94.6%

Faculty of Technology, University of Delhi

Bachelor of Technology in Electronics and communication Engineering

Minors: VLSI Design

Additional Courses

- Image Processing in Matlab
- Financial Literacy

Projects

NOAA Satellite Signal Reception System

Jan 2025 – Ongoing

- Designed a custom antenna to receive NOAA satellite signals using WEB-SDR.
- Researched and implemented remote SDR solutions to minimize hardware costs.
- Processed received satellite images using software-based demodulation and decoding techniques.
- Conducted signal analysis to optimize reception quality and reduce interference.
- Experimenting with different antenna designs and materials to improve signal strength.
- Documenting the process and findings to contribute to open-source satellite reception communities.

Skills

- **Simulation & Circuit Design:** MATLAB, Silvaco TCAD, LTSpice, Multisim
- **VLSI & Semiconductor Analysis:** MOSFET and CMOS
- **device modeling, circuit design, and process simulation**
- **Programming & Scripting:** Python, C
- **Tools & Environments:** Linux, Git, Visual Studio Code

Extracurricular

Astronomy and Space Science Community Involvement

Ongoing

- Active member of online astronomy forums and satellite tracking groups.

VLSI and Semiconductor Learning Initiatives

Ongoing

- Engaged in self-paced coursework on VLSI and CMOS circuit design through online platforms.
- Explored industry trends in semiconductor manufacturing and chip design.

Chess Enthusiast and Strategic Thinking

Ongoing

- Studied advanced chess theory, including openings, endgames, and positional play.
- Regularly played and analyzed games to improve strategic thinking and problem-solving.